ADITHYA BHASKAR

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Education

2023-	Ph.D. in Computer Science, Princeton University, USA *Advised by Prof. Danqi Chen* (Specialization : Natural Language Processing) **Research interests :** AI for science, LM post-training, Interpretability	4.00/4.00
2019-23	Bachelor of Technology in Computer Science and Engineering (Honors), IIT Bombay, India	9.67/10.00
2017-19	High School, Central Board of Secondary Education, India	97.2/100.0
2017	Senior Secondary School, Central Board of Secondary Education, India	98.0/100.0

Publications

2026	Language Models That Think, Chat Better, **Adithya Bhaskar***, Xi Ye*, and Danqi Chen	*COLM 2026*
2026	Continual Memorization of Factoids in Language Models, Howard Chen, Jiayi Geng, **Adithya Bhaskar**, Dan Friedman, and Danqi Chen	*TMLR, March 2026*
2025	Unintentional Unalignment : Likelihood Displacement in Direct Preference Optimization, Noam Razin, Sadhika Malladi, **Adithya Bhaskar**, Danqi Chen, Sanjeev Arora, and Boris Hanin	*ICLR 2025*
2024	Finding Transformer Circuits With Edge Pruning, **Adithya Bhaskar**, Alexander Wettig, Dan Friedman, and Danqi Chen	*NeurIPS 2024* **(Spotlight)**
2024	The Heuristic Core : Understanding Subnetwork Generalization in Pretrained Language Models, **Adithya Bhaskar**, Dan Friedman, and Danqi Chen	*ACL 2024* **(Oral)**
2023	Benchmarking and Improving Text-to-SQL Generation under Ambiguity, **Adithya Bhaskar***, Tushar Tomar*, Ashutosh Sathe, and Sunita Sarawagi	*EMNLP 2023*
2023	Prompted Opinion Summarization with GPT-3.5, **Adithya Bhaskar**, Alexander R. Fabbri, and Greg Durrett	*ACL 2023 (Findings)*
2023	Performance Bounds for LASSO under Multiplicative Noise : Applications to Pooled RT-PCR Testing, Richeek Das, Aaron Jerry Ninan, **Adithya Bhaskar**, and Ajit Rajwade	*Signal Processing, Vol. 214*

Preprints

2026	Language Models that Play Chess and Explain Their Moves, **Adithya Bhaskar***, Jeffrey Cheng*, and Danqi Chen	*arXiv preprint, arXiv:2610.03695*
2025	Cache Me If You Can : How Many KVs Do You Need for Effective Long-Context LMs ?, **Adithya Bhaskar***, Alexander Wettig*, Tianyu Gao, Yihe Dong, and Danqi Chen	*arXiv preprint, arXiv:2506.17121*
2025	Extracting Rule-based Descriptions of Attention Features in Transformers, Dan Friedman, **Adithya Bhaskar**, Alexander Wettig, and Danqi Chen	*arXiv preprint, arXiv:2510.18148*
2024	Improving Language Understanding from Screenshots, Tianyu Gao, Zirui Wang, **Adithya Bhaskar**, and Danqi Chen	*arXiv preprint arXiv:2402.14073*

Invited Talks

April 2024	The Heuristic Core : Understanding Subnetwork Generalization in Pretrained Language Models *Host : Mathew Monfort*	*Amazon AWS*

Scholastic Achievements

2024 Recipient of the **Hisashi and Masae Kobayashi *67 Fellowship**.
2023 Recipient of the **Thomas Dooie Class of 1974 Research Award**.
2019 **All India Rank 18** in JEE Advanced 2019 among **240 thousand** candidates.
2019 **All India Rank 114** in JEE Mains 2019 among **1.1 million** candidates.
2018 Secured a position in the **top 39 ranks** in the **Indian National Physics Olympiad** and was **invited to the Orientation-cum-Selection-Camp in Physics** held in May-June 2018.
2018 Secured a position in the **top 49 ranks** in the **Indian National Chemistry Olympiad** and was **invited to the Orientation-cum-Selection-Camp in Chemistry** held in May-June 2018.
2016 Among the **39 students** to clear the **Indian National Mathematical Olympiad**, becoming **one of the youngest** to **ever** be invited to the Orientation-cum-Selection-Camp in Mathematics **aged 14**.

Experience

UT Austin
Summer 2022

Research Intern, Natural Language Processing, USA
Advisor : Prof. Greg Durrett
Very Large Language Models for Multi-Document Summarization

- Developed metrics to measure **factuality**, **faithfulness** and **specificity** (whether it is correct, prefers major viewpoints, and is not too generic) for a summary of multi-document text such as hotel reviews.
- Utilized the above along with an n-gram abstractiveness metric to benchmark **GPT-3.5**, and showed that simple hierarchical summarization of large text performs poorly on faithfulness and specificity.
- Investigated various **pre-clustering** and **pre-summarization** methods and illustrated that pre-summarization with a pretrained keyword-based extractive model improves correctness, faithfulness and specificity, while only marginally affecting abstractiveness.

Uppsala University
Summer 2021

Research Intern, Formal Verification,
Advisor : Prof. Parosh Abdulla
Model Checking for Programs Running under the ARMv8 Memory Model

- Developed a **model** and **simulator** for programs running under the ARMv8 memory model, and demonstrated the equivalence of the model to the ARM specification using **7500+** litmus tests.
- Used **Context Bounded Model Checking** to perform **State Reachability Analysis** for programs under the ARMv8 memory model, achieving **up to an order of magnitude** of speedup over existing checkers.

Teaching

Spring 2025 Graduate Teaching Assistant, *COS 484 : Natural Language Processing* — Princeton University
Instructors : Danqi Chen, Vikram Ramaswamy, and Tri Dao

Fall 2024 Graduate Teaching Assistant, *COS 597R : Deep Dive into Large Language Models* — Princeton University
Instructors : Danqi Chen and Sanjeev Arora

Service

2026	ICLR 2026	Reviewer
2026	COLM 2026	Reviewer
2025	NeurIPS 2025	Reviewer
2025	ICML 2025 MOSS Workshop	Reviewer
2024	NeurIPS 2024 ATTRIB Workshop	Reviewer

Programming Languages

Proficient in Python, C++, C, Bash, Golang, SQL **Familiar with** MATLAB, Java, Haskell